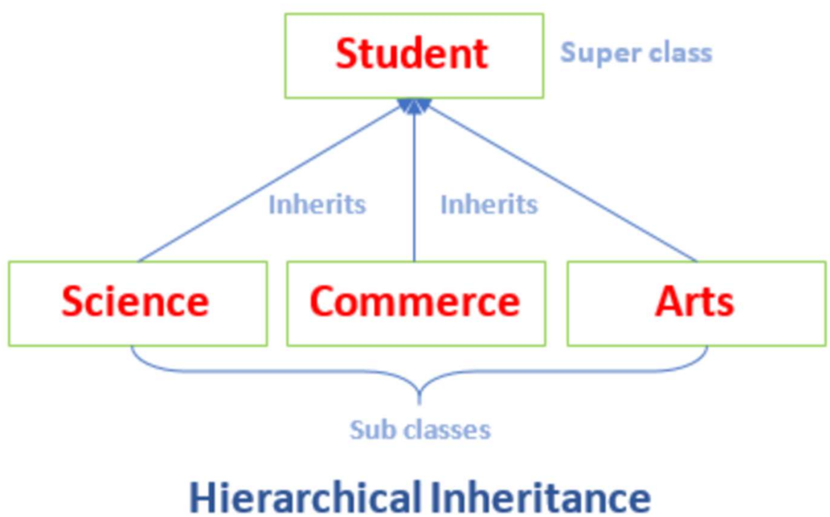
Types of Inheritance

Hierarchical Inheritance :-

- Hierarchical inheritance is a type of inheritance in which a **parent class** can have **multiple child classes**.
- In other words, it is a type of inheritance in which a **parent class serves as a base for multiple child classes**, and **each child class** inherits properties and methods from the parent class.



Key Points :-

- **Parent class**: Contains common properties/methods.
- **Multiple child classes** inherit the parent independently.
- Each child can have **its own unique methods** along with parent methods.

Implementation of Hierarchical Inheritance :-

Input :-

```java
package Oops;

public class hi_parent {

    void p()
    {
        System.out.println("Parent ");
    }
}

```

```java
package Oops;

public class hi_child2 extends hi_parent{

    void c2()
    {
        System.out.println("Child 2");
    }
}
```

```java
package Oops;

public class hi_child1 extends hi_parent {

void c1()
{
    System.out.println("Child 1");
}

public static void main(String[] args) {

    hi_child1 ob=new hi_child1();
    ob.p();
    ob.c1();

    hi_child2 ob2 = new hi_child2();
    ob2.c2();

}
}
```

Output :-

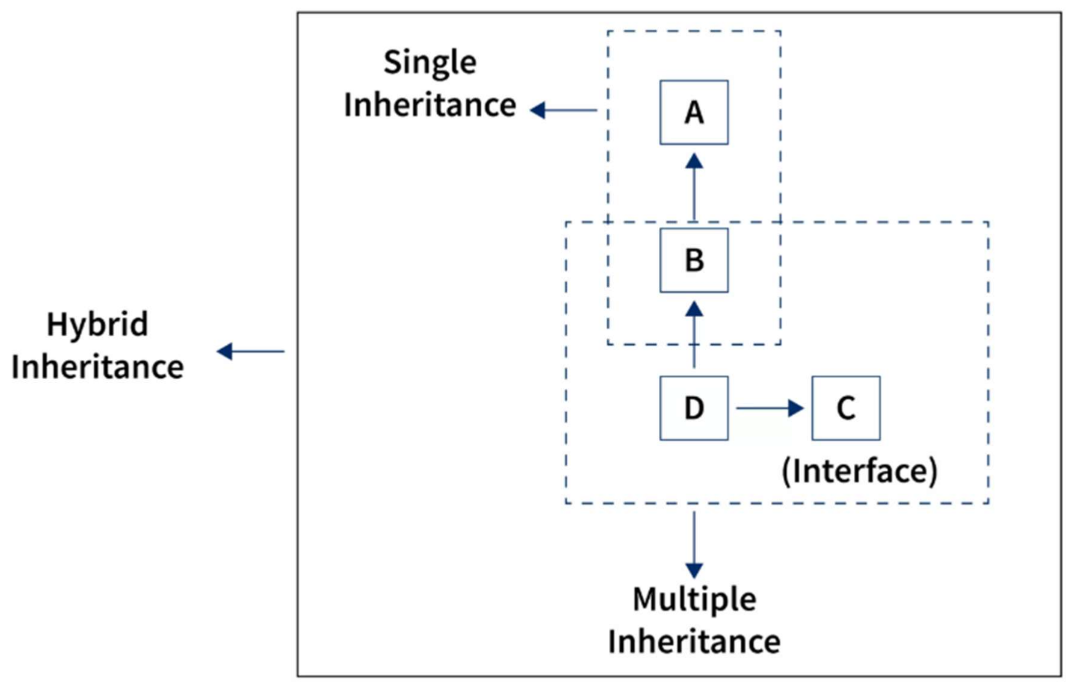
```
Parent
Child 1
Child 2
```

Hybrid Inheritance :-

- Hybrid Inheritance is a **combination of two or more types** of inheritance (like **single**, **multilevel** and **hierarchical.**) in a **single program.**

For Example:-

- A mix of **Hierarchical** + **Multilevel**
- A mix of **Single** + **Hierarchical** etc.



Key Points :-

- Hybrid inheritance is a **combination** of different inheritance types.
- **Not supported with only classes** (to avoid Diamond Problem).
- Can be **achieved using interfaces**.

Implementation of Hybrid Inheritance :-

Input :-

```java
package Oops;

class A
{
    void displayA()
    {
        System.out.println("Class A");
    }
}

class B extends A
{
    void displayB()
    {
        System.out.println("Class B");
    }
}

class C extends A
{
    void displayC()
    {
        System.out.println("Class C");
    }
}

class D extends B
{
    void displayD()
    {
        System.out.println("Class D");
    }
}

public class hy_college {

    public static void main(String[] args) {

        D ob = new D();
        ob.displayA();
        ob.displayB();
        ob.displayD();

        C ob1 = new C();
        ob1.displayA();
        ob1.displayC();
    }
}
```

Output :-

```
Class A
Class B
Class D
Class A
Class C
```